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# Applied Cyber Threat Analysis – Network Traffic Analysis Lab

- This lab was conducted within the Virginia Cyber Range Environment.
  - Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
  - Cheche Agada. (2024). Email Header Analysis Lab IT-462: Applied Cyber Threat Analysis, George Mason University
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## Introduction

Network traffic analysis is an important part of digital investigations. Network traffic may contain evidence of malicious activity or network device failure/malfunction. In this lab, you will investigate a packet capture (PCAP) file using GUI tools as well as command line tools.

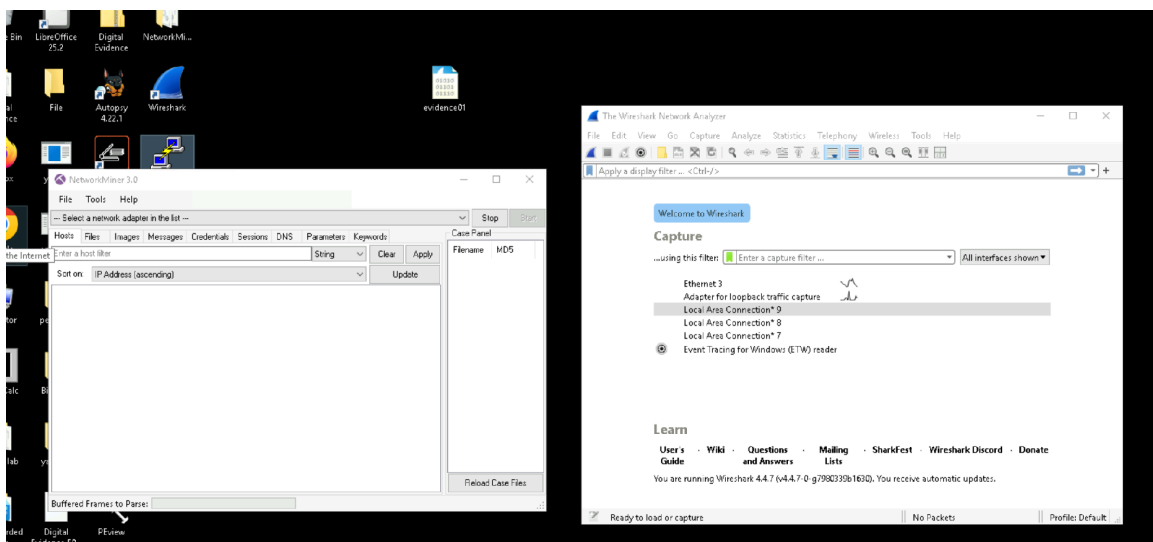
## Objective

Conduct an analysis of a PCAP file.

## Part 1

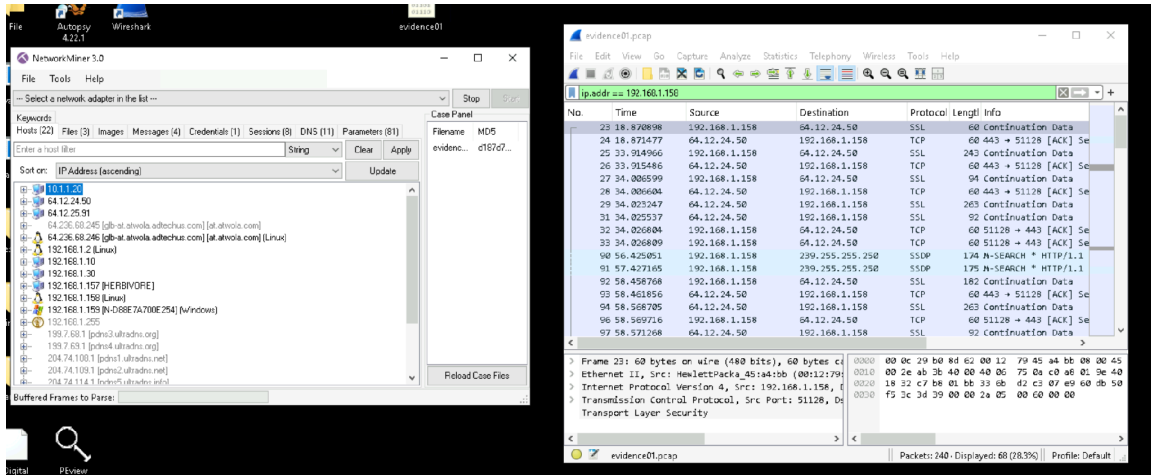
### Task 1: Install Wireshark and NetworkMiner

- Download and install Wireshark (<https://www.wireshark.org/>)
- Download NetworkMiner (<https://www.netresec.com/?page=NetworkMiner>)



## Task 2: Analyze a Packet Capture (PCAP) file (GUI Tools)

- Download the file named 'evidence01.pcap' (Evidence File)
- Open the file in Wireshark as well as NetworkMiner.
- Use specific evidence (i.e. frame numbers, screenshots, etc.) from the PCAP file to answer the questions in exercise 1.

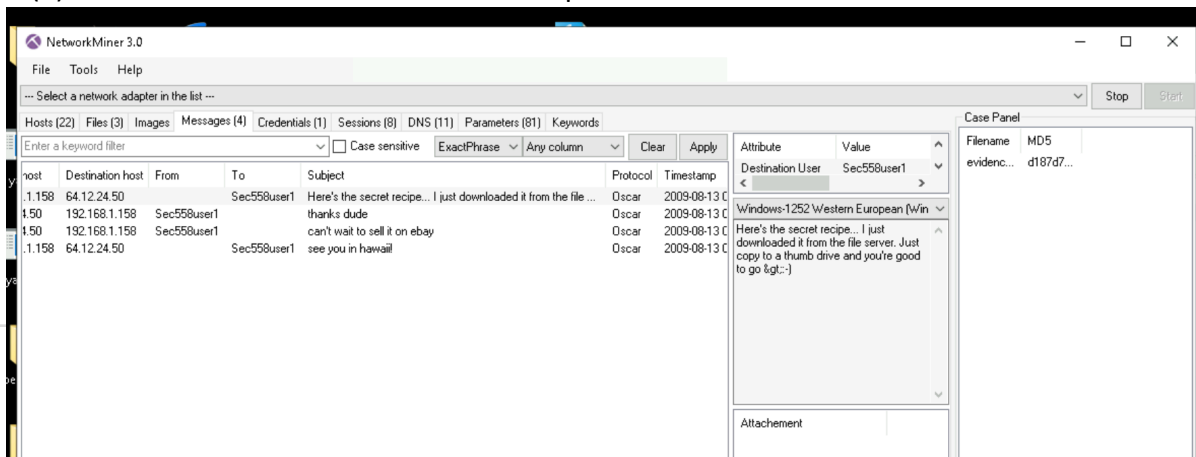


## Exercise 1

**Scenario:** Anarchy-R-U's, Inc. suspects that one of their employees, Ann Dercover, is really a secret agent working for their competitor. Ann has access to the company's prized asset, the secret recipe. Security staff are worried that Ann may try to leak the company's secret recipe. Security staff have been monitoring Ann's activity for some time, but haven't found anything suspicious— until now. Today an unexpected laptop appeared on the company wireless network. Staff hypothesize it may have been someone in the parking lot, because no strangers were seen in the building. Ann's computer, (192.168.1.158) sent Instant Messages (IMs) over the wireless network to this computer. The rogue laptop disappeared shortly thereafter. "We have a packet capture of the activity," said security staff, "but we can't figure out what's going on. Can you help?"

You are the forensic investigator. Your mission is to figure out who Ann was IM-ing, what she sent, and recover evidence including:

- (1) What is the name of Ann's IM buddy? **Sec558user1 – 64.12.24.50**
- (2) What was the first comment in the captured IM conversation?



“ Here’s the secret recipe... I just downloaded it from the file server. Just copy to a thumb drive and you’re good to go &gt;-) ”

(3) What is the name of the file Ann transferred? **recipe.docx**

| --- Select a network adapter in the list ---                     |                           |           |          |               |
|------------------------------------------------------------------|---------------------------|-----------|----------|---------------|
| Hosts (22) Files (3) Images Messages (4) Credentials (1) Session |                           |           |          |               |
| Enter a keyword filter                                           |                           |           |          |               |
| Frame nr.                                                        | Filename                  | Extension | Size     | Source        |
| 112                                                              | recipe.docx               | docx      | 12 008 B | 192.168.1.100 |
| 230                                                              | size=120x90;noperf=1.html | html      | 375 B    | 64.230.1.1    |
| 233                                                              | size=120x90;noperf=1.js   | js        | 335 B    | 64.230.1.1    |

(4) What is the magic number of the file you want to extract (first four bytes)?

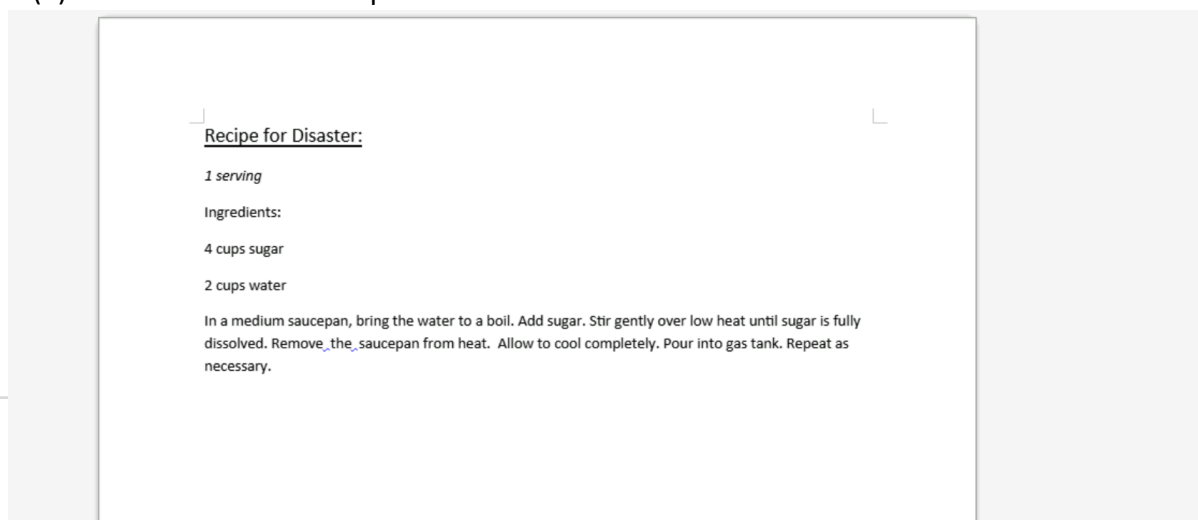
**504B0304**



(5) What is the MD5sum of the file? **8350582774E1d4dbe1d61d64c89e0ea1**

|        |                                                                |
|--------|----------------------------------------------------------------|
| Name   | recipe.docx                                                    |
| MD5    | 8350582774e1d4dbe1d61d64c89e0ea1                               |
| SHA1   | 11745854a7f8cd0c513dbaa695e84fde9fa0e581                       |
| SHA256 | f0f74a982a814640aedaa5fd6542ac810e8c5e257552bcc024a5c8083      |
| Path   | C:\Users\student\AppData\Local\NetworkMiner\AssembledFiles\192 |
| Size   | 12008                                                          |

(6) What is the secret recipe?



### Recipe for Disaster:

1 serving

Ingredients:

4 cups sugar

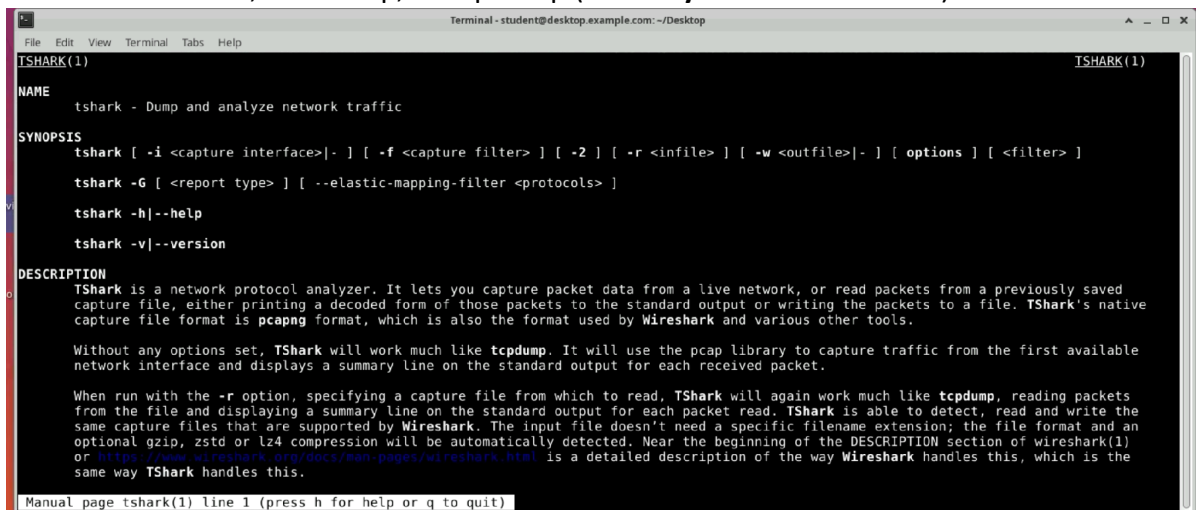
2 cups water

In a medium saucepan, bring the water to a boil. Add sugar. Stir gently over low heat until sugar is fully dissolved. Remove the saucepan from heat. Allow to cool completely. Pour into gas tank. Repeat as necessary.

## Part 2

### Task 1: Command Line Tools

- You may complete this portion of the lab using a Windows or Linux command line.
- Install TShark, WinDump, or Tcpdump (**You only need one of them**).



```
Terminal - student@desktop.example.com: ~/Desktop
TSHARK(1)
NAME
    tshark - Dump and analyze network traffic
SYNOPSIS
    tshark [ -i <capture interface>|- ] [ -f <capture filter> ] [ -2 ] [ -r <infile> ] [ -w <outfile>|- ] [ options ] [ <filter> ]
    tshark -G [ <report type> ] [ --elastic-mapping-filter <protocols> ]
    tshark -h|--help
    tshark -v|--version
DESCRIPTION
    TShark is a network protocol analyzer. It lets you capture packet data from a live network, or read packets from a previously saved
    capture file, either printing a decoded form of those packets to the standard output or writing the packets to a file. TShark's native
    capture file format is pcapng format, which is also the format used by Wireshark and various other tools.

    Without any options set, TShark will work much like tcpdump. It will use the pcap library to capture traffic from the first available
    network interface and displays a summary line on the standard output for each received packet.

    When run with the -r option, specifying a capture file from which to read, TShark will again work much like tcpdump, reading packets
    from the file and displaying a summary line on the standard output for each packet read. TShark is able to detect, read and write the
    same capture files that are supported by Wireshark. The input file doesn't need a specific filename extension; the file format and an
    optional gzip, zstd or lz4 compression will be automatically detected. Near the beginning of the DESCRIPTION section of wireshark(1)
    or https://www.wireshark.org/docs/man-pages/wireshark.html is a detailed description of the way Wireshark handles this, which is the
    same way TShark handles this.

Manual page tshark(1) line 1 (press h for help or q to quit)
```

#### NOTE:

- **TShark** is the command line version of Wireshark and is installed with Wireshark. To run TShark on Windows, there are 2 options. You can navigate to the Wireshark folder on your command line and run TShark from there each time (not convenient), or you may add the Wireshark folder path to your system/environment variables so you can run TShark from any location on the command line (recommended).
- **WinDump** works on Windows while **Tcpdump** works on Linux.
- See the resource links below for help with installing these tools.

### Task 2: Analyze a Packet Capture (PCAP) file (Command Line Tools)

## Exercise 2

(1) What is the name of Ann's IM buddy? **Sec558user1**

```
05:58:34.156145 IP 64.12.24.50.https > ip-192-168-1-158.ec2.internal.51128: Flags [P.], seq 1463:1501, ack 523, win 64240, length 38
0x0000: 4500 004e b499 0000 7f06 6c8c 400c 1832 E..N.....l@..2
0x0010: c0a8 019e 01bb c7b8 07e9 6691 336b d4ce .....f.3k...
0x0020: 5018 faf0 3d13 0000 2a02 56e0 0020 0004 P...=...*.V.....
0x0030: 000c 0000 0000 0049 3530 3838 3439 3600 .....I5088496.
0x0040: 0001 0b53 6563 3535 3875 7365 7231 ....Sec558user1..
05:58:34.184973 IP ip-192-168-1-158.ec2.internal.51128 > 64.12.24.50.https: Flags [P.], seq 6:195, ack 1, win 62780, length 0
0x0000: 4500 0028 ab4d 4000 4006 74fe c0a8 019e E..(M@.t.....
0x0010: 400c 1832 c7b8 01bb 336b d2c9 07e9 60db @..2....3k....
0x0020: 5018 f53c d08c 0000 2a02 0061 00b7 0004 P..<....*.a....
0x0030: 0006 0000 0000 0045 3436 3238 3737 3800 .....E4628778.
0x0040: 0001 0b53 6563 3535 3875 7365 7231 0002 ....Sec558user1..
0x0050: 000f 0501 0004 0101 0102 0101 0083 0000 .....
0x0060: 0000 4865 7265 2773 2074 6865 2073 6563 ..Here's.the.sec
0x0070: 7265 7420 7265 6369 7065 2e2e 2e20 4920 ret.recipe...I.
0x0080: 6a75 7374 2064 6f77 6e6c 6f61 6465 6420 just.downloaded.
0x0090: 6974 2066 726f 6d20 7468 6520 6669 6c65 it.from.the.file
0x00a0: 2073 6572 7665 722e 204a 7573 7420 636f .server..Just.co
0x00b0: 7079 2074 6f20 6120 7468 756d 6220 6472 py.to.a.thumb.dr
0x00c0: 6976 6520 616e 6420 796f 7527 7265 2067 ive.and.you're.g
0x00d0: 6f6f 6420 746f 2067 6f20 2667 743b 3a2d ood.to.go.&gt;:-
0x00e0: 2900 0300 00 .....
).....
```

(2) What was the first comment in the captured IM conversation?

```
05:57:37.066461 IP ip-192-168-1-158.ec2.internal.51128 > 64.12.24.50.https: Flags [P.], seq 6:195, ack 1, win 62780, length 189
0x0000: 4500 00e5 ab3c 4000 4006 7452 c0a8 019e E....<@.tR....
0x0010: 400c 1832 c7b8 01bb 336b d2c9 07e9 60db @..2....3k....
0x0020: 5018 f53c d08c 0000 2a02 0061 00b7 0004 P..<....*.a....
0x0030: 0006 0000 0000 0045 3436 3238 3737 3800 .....E4628778.
0x0040: 0001 0b53 6563 3535 3875 7365 7231 0002 ....Sec558user1..
0x0050: 000f 0501 0004 0101 0102 0101 0083 0000 .....
0x0060: 0000 4865 7265 2773 2074 6865 2073 6563 ..Here's.the.sec
0x0070: 7265 7420 7265 6369 7065 2e2e 2e20 4920 ret.recipe...I.
0x0080: 6a75 7374 2064 6f77 6e6c 6f61 6465 6420 just.downloaded.
0x0090: 6974 2066 726f 6d20 7468 6520 6669 6c65 it.from.the.file
0x00a0: 2073 6572 7665 722e 204a 7573 7420 636f .server..Just.co
0x00b0: 7079 2074 6f20 6120 7468 756d 6220 6472 py.to.a.thumb.dr
0x00c0: 6976 6520 616e 6420 796f 7527 7265 2067 ive.and.you're.g
0x00d0: 6f6f 6420 746f 2067 6f20 2667 743b 3a2d ood.to.go.&gt;:-
0x00e0: 2900 0300 00 .....
).....
```

Here's the secret recipe... I just downloaded it from the file server. Just copy to a thumb drive and you're good to go &gt;:-)"

(3) What is the name of the file Ann transferred?

```
05:58:01.610263 IP ip-192-168-1-158.ec2.internal.51128 > 64.12.24.50.https: Flags [P.], seq 235:363, ack 248, win 62742, length 0
0x0000: 4500 00a8 ab40 4000 4006 748b c0a8 019e E....@.@.t....
0x0010: 400c 1832 c7b8 01bb 336b d3ae 07e9 61d2 @..2....3k....a.
0x0020: 5018 f516 df0b 0000 2a02 0063 007a 0004 P.....*.c.z..
0x0030: 0006 0000 0000 0047 3731 3734 3634 3700 .....G7174647.
0x0040: 0002 0b53 6563 3535 3875 7365 7231 0003 ....Sec558user1..
0x0050: 0000 0005 0052 0000 3731 3734 3634 3700 .....R..7174647.
0x0060: 0946 1343 4c7f 11d1 8222 4445 5354 0000 .F.CL...."DEST..
0x0070: 0002 0004 c0a8 019e 0003 0004 c0a8 019e .....
0x0080: 0005 0002 144e 000a 0002 0001 000f 0000 .....F.....
0x0090: 2711 0014 0001 0001 0000 2ee8 7265 6369 '.....reci
0x00a0: 7065 2e64 6f63 7800 pe.docx.
05:58:01.613351 IP 64.12.24.50.https > ip-192-168-1-158.ec2.internal.51128: Flags [P.], seq 363, win 64240, length 0
0x0000: 4500 0028 b46f 0000 7f06 6cdc 400c 1832 E..(o.....l@..2
0x0010: c0a8 019e 01bb c7b8 07e9 61d2 336b d42e .....a.3k...
--More--
```

**recipe.docx**

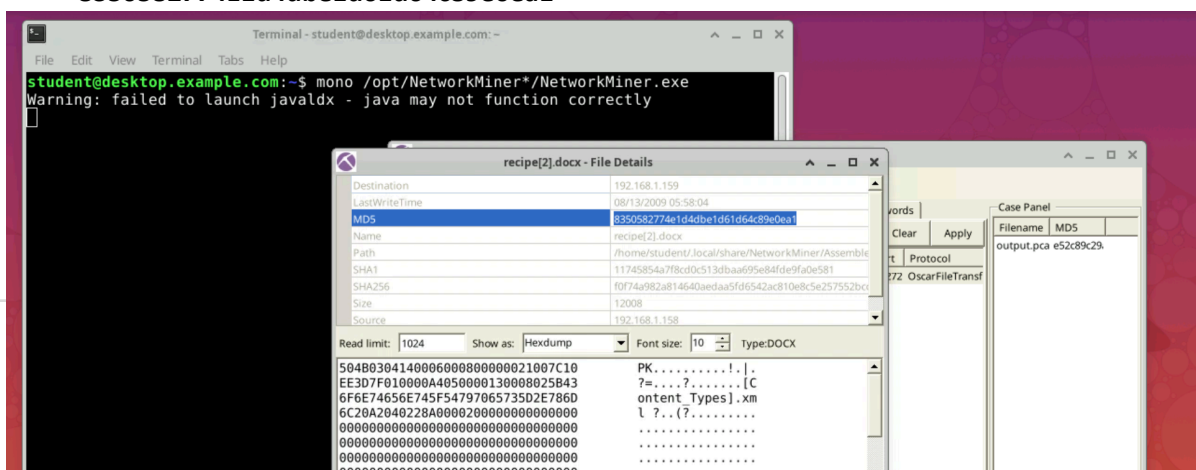
(4) What is the magic number of the file you want to extract (first four bytes)?

```
0100 75 00 00 00 00 00 00 00 00 00 00 00 00 00 00 x.....
0110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
student@desktop.example.com:~/Desktop$ tshark -r evidence01.pcap -Y "data.data[0:4]==50.4b.03.04" -x|grep --color=always -b "50 4b 03 04"
219:0030 19 20 10 54 00 00 50 4b 03 04 14 00 06 00 08 00 .T.PK.....
student@desktop.example.com:~/Desktop$ tshark -r evidence01.pcap -Y "data.data[0:4]==50.4b.03.04" -x|grep --color=always -b "50 4b 03 04"
```

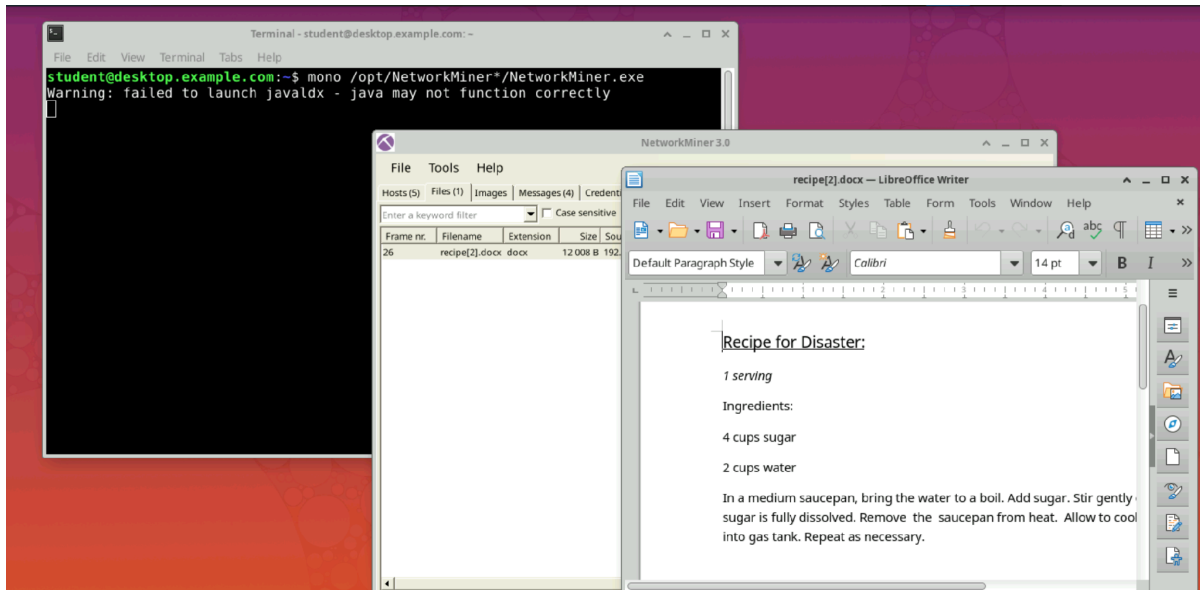
**504B0304**

(5) What is the MD5sum of the file?

**8350582774E1d4dbe1d61d64c89e0ea1**



(6) What is the secret recipe?



**Recipe for Disaster:**

**1 serving**

**Ingredients:**

**4 cups sugar**

**2 cups water**

**In a medium saucepan, bring the water to a boil. Add sugar. Stir gently over low heat until sugar is fully dissolved. Remove the saucepan from heat. Allow to cool completely. Pour into gas tank. Repeat as necessary.**